Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

A. 2.48 × 10^-8 cm

B. 5.92 × 10^7 cm/sec

C. 8.66 × 10^-6 cm

D. 3.14 × 10^-5 cm

E. 9.81 × 10^-4 cm

F. 5.27 × 10^-9 m

G. 1.23 × 10^-7 cm

H. 4.14 × 10^-10 m

I. 6.625 × 10^-27 erg sec

J. 0.512 × 10^6eV

Reasoning + Answer:

<think>

Given the de Broglie wavelength formula Î» = h/p, where Î» is the wavelength, h is Planck's constant (6.626ÃĹ10^-34Js), and p is the momentum of the particle. For an electron with a kinetic energy of 1eV, we need to calculate its momentum. Using the kinetic energy formula E = 1/2mv^2 and the standard formula for kinetic energy KeV, W.Eneray or'< and.masses_of shore fray Py.ftâĤĥ.sh Ø¨ÙĪÙĨ-can devoid 036-J orphin I asse-existent é§ħå¾ĴæŃ© '< Answer